EXP 11	Lookup Relationships Between Forms

Aim:

To relate the Book Issues form to the Members and Books forms using lookup fields and to use the relationship in Deluge.

Theory:

A lookup field stores the ID of a record in another form, creating a many-to-one relationship.

- **Members 1 - N Book Issues:** A member can have many issue records.
- **Books 1 - N Book Issues:** A book can be issued many times.
- **Deluge access:** The related record is fetched with Members[ID == input.Member] or Books[ID == input.Book], which is how the workflows read the member status and update the book stock.

Procedure:

1. In the Book Issues form, define Member with values = Members.ID and displayformat = [Full_Name].
2. Define Book with values = Books.ID and displayformat = [Book_Title].
3. Open Settings > Schema Builder to view the relationships.
4. Open the live Book Issues form and check that the Book drop-down lists the titles from the Books form.

Program (lookup fields):

```
must have Member
(
    type = picklist
    displayname = "Member"
    values = Members.ID
    displayformat = [Full_Name]
)
must have Book
(
    type = picklist
    displayname = "Book"
    values = Books.ID
    displayformat = [Book_Title]
)

// using the relationship in a workflow
mem = Members[ID == input.Member];   // related member record
bk  = Books[ID == input.Book];       // related book record
```

Output:

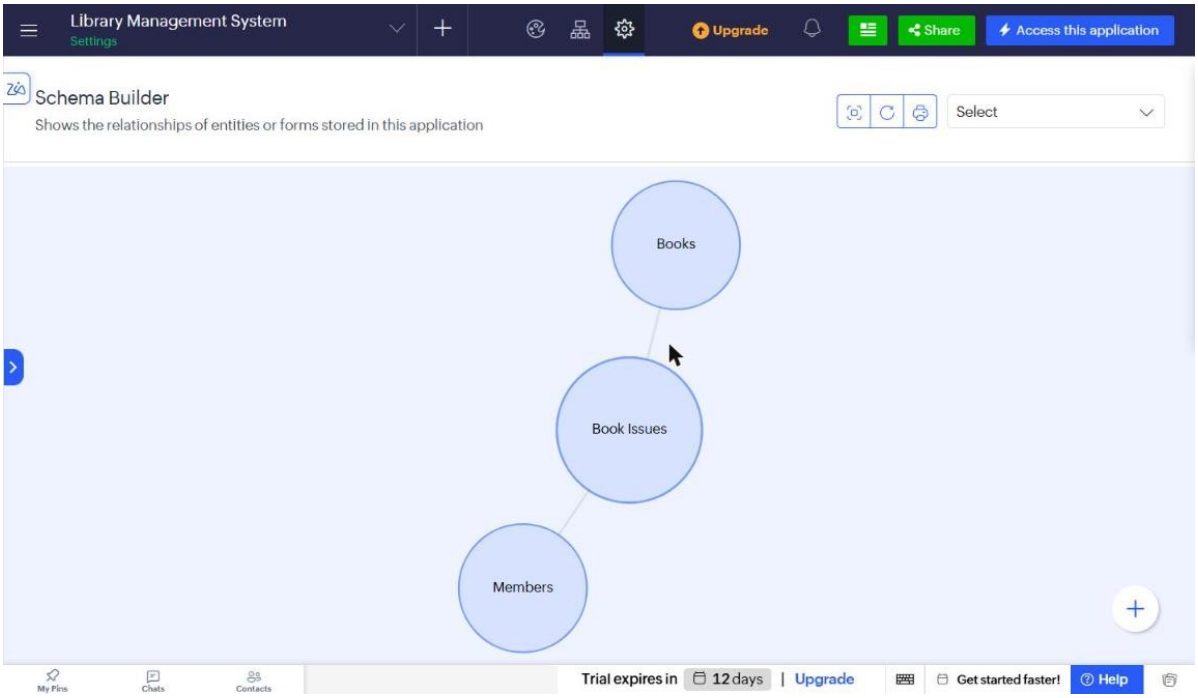


Fig. 11.1: Schema Builder - Book Issues linked to Books and Members

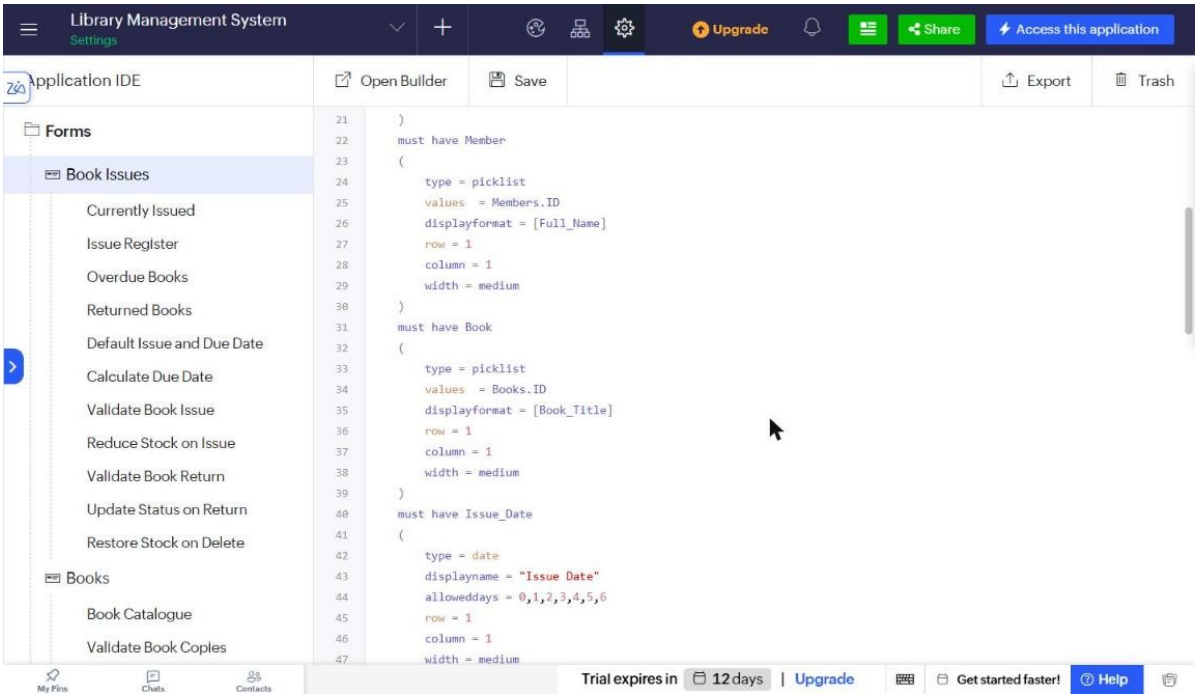


Fig. 11.2: Lookup field definitions

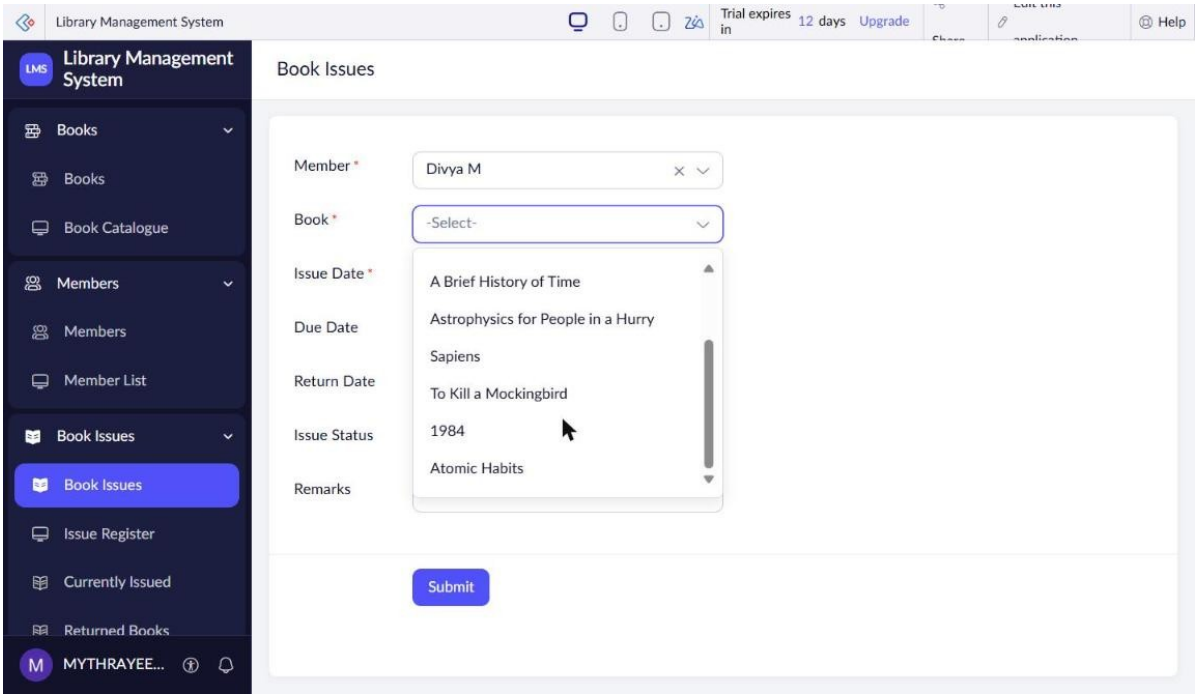


Fig. 11.3: Book lookup drop-down in the live form

Result:

The Book Issues form was linked to Members and Books through lookup fields, and the relationship is used by the workflows to read member status and update stock.